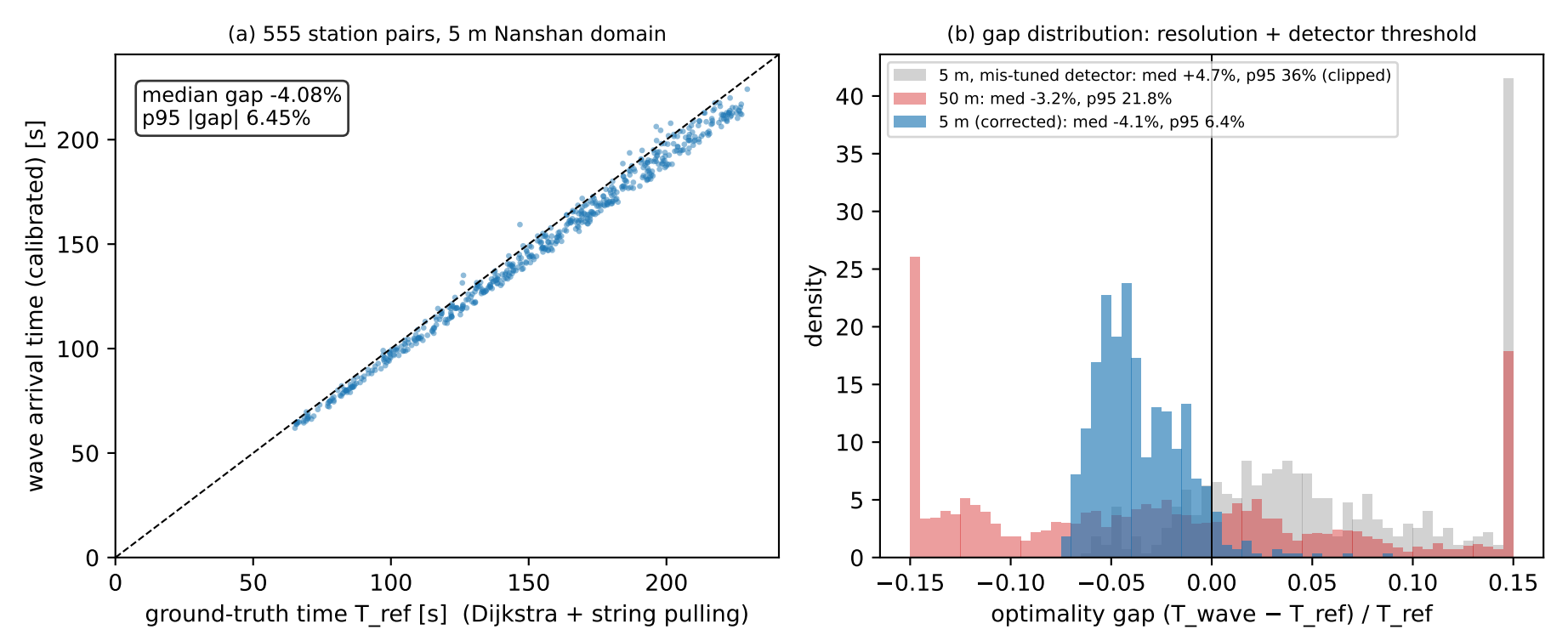
(a) 555 station pairs, 5 m Nanshan domain
median gap -4.08%
p95 |gap| 6.45%
wave arrival time (calibrated) [s]
ground-truth time T_ref [s] (Dijkstra + string pulling)
0
50
100
150
200
(b) gap distribution: resolution + detector threshold
5 m, mis-tuned detector: med +4.7%, p95 36% (clipped)
50 m: med -3.2%, p95 21.8%
5 m (corrected): med -4.1%, p95 6.4%
density
optimality gap (T_wave − T_ref) / T_ref
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0
5
10
15
20
25
30
35
40